Developing detrital methods to investigate source-sink relationships of microplastics in the Columbia River watershed

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ABSTRACT

Microplastics (MPs) are ubiquitous and potentially dangerous particles transported primarily by streams. However, little work has been done to rigorously define MP transport within these waterways. Although research about MPs is rapidly growing in popularity, most studies have presented findings about MP assemblages in waterways, but have not quantitatively compared these assemblages to each other. We hypothesize (1) that different streams have distinct MP fingerprints, (2) that downstream MP fingerprints represent the admixture of upstream/tributary sources, and (3) that downstream of a catchment, the catchment's land use is recorded in MP assemblages. In this project, we will define MP presence and transport in the Columbia River watershed (CRW). This process will be done by applying established quantitative comparison methods used in geologic sediment provenance studies to MP samples collected in the tributaries and main stem of the Columbia River. This new application of established methods will allow us to rigorously define MP transport in the CRW and beyond.

INTRODUCTION

Microplastics

Microplastics (MPs) are particles of plastic with a diameter of less than 5 mm. Plastics are a group of synthetic polymers derived from petroleum, and their global production has increased substantially in recent decades (Rangel-Buitrago et al., 2026). Common varieties of plastics include acrylonitrile styrene polymers (SAN), ethylene-vinyl acetate (EVA), nylon, polycarbonate,

polyethylene (PE), polyethylene/butylene terephthalate (PET/PBT), polymethyl methacrylate (PMMA), polypropylene (PP), polystyrene (PS), polyurethane (PUR), and polyvinyl chloride (PVC). Original shape, weathering, and other factors can determine the shape of MPs (Rangel-Buitrago et al., 2026).

Most plastics are resistant to chemical weathering, but they readily weather mechanically (Rangel-Buitrago et al., 2026). This process of mechanical weathering produces MPs, which are persistent in the environment, but almost undetectable to the naked eye. Although MPs have been a known source of pollution for several decades (e.g., Ray, 1973; Johnston, 1976), they have now become a popular focus of research (Thompson et al., 2024). Recently, they have been linked to several problems in humans such as infertility and cancer (Dzierżyński et al., 2024). Recent studies have shown that they are ubiquitous in oceans and waterways around the world (Rangel-Buitrago et al., 2026). In this study, we will quantify MPs transport in waterways.

Study Area

The area of focus in this study is the Columbia River watershed (CRW) located in the northwestern United States and southwestern Canada (Figure 1). The CRW drains an area of approximately 1,300,000 km^2, spanning areas in Idaho, Wyoming, Oregon, Washington, Utah, Nevada, and British Columbia. Most of the watershed area lies in the interior parts of the Pacific Northwest (Lehner and Grill, 2013). In this region, streamflow is predominantly produced by springtime snowmelt (Hamlet and Lettenmeier, 1999). As the snow melts each spring, particles from across the landscape are transported into streams.

The CRW is a vital resource for around 8.3 million people. Its waterways provide food, water, power, work, and recreation to those living both within and outside of the watershed. The watershed is of ongoing importance to several indigenous groups including, but not limited to, the Shoshone, Paiute, Nimiipuu, Sahaptin, and Salish peoples (Sampson, 2015). Public, private, and tribal entities are involved in monitoring streams in the CRW and mitigating issues including contamination from

pesticides, erosion from mine tailings, and pollution from septic and stormwater systems (Environmental Protection Agency, 2025).

The CRW drains a variety of land uses including urban/industrial (0.6%), agricultural (7.6%), and wilderness areas (90.3%) (Figure 2a, Figure 2b). Some of the urban centers include Portland, Salem, and Bend in Oregon, Spokane, Yakima, and the Tri-Cities area in Washington, and the Treasure Valley and Idaho Falls in Idaho. Major agricultural centers include the Palouse, the Snake River Plain, and the Willamette Valley. The CRW includes vast wilderness areas such as the Canadian Rockies, the Idaho Batholith, the Eastern Cascades, the Tetons, and coastal ranges within Oregon and Washington.

Microplastic Sources

Different land uses and ground covers may generate different amounts and assemblages of MPs worldwide. Agricultural areas have high MP contamination that has been linked to the application of sewage sludge and plastic mulch. Even remote areas have a baseline contamination level from eolian MPs (Büks and Kaupenjohann, 2020). MPs from soils can also enter waterways, where abundance is driven partially by land use, population density, per-capita GDP, elevation, sediment load, and water temperature (Zhang et al 2024). Although no such data exists for the CRW already, these studies may prove useful in thinking about MPs transport. Because multiple land uses can co-occur within a watershed, attributing MPs in downstream areas to specific upstream sources requires development of quantitative approaches.

Sources, Sinks, and Provenance

Sediment sources are traditionally defined as rocks that are being uplifted, weathered, and eroded (Schumm, 1977). The majority of sediment produced is transported by streams (cf., ice, wind), stored, and eventually deposited in sedimentary basins (i.e., sinks) (Allen, 2017). Traditionally, methods like compositional analysis, paleocurrent analysis, sandstone petrography/microscopy, and grain size analysis have helped to determine sediment provenance. These methods have been extended and integrated into quantitative analysis of geochronology and geochemical data. More recently,

detrital zircon (DZ) geochronology has played an important role in determining sediment provenance. On the most basic level, this subfield uses radiometric dating to obtain the ages of rocks. The DZ geochronology subfield has seen the development and application of several new quantitative tools used to quantify sediment transport. Of these new quantitative tools, multidimensional scaling and mixture modeling are some of the most interpretive and can provide insights into sediment provenance across large systems (Vermeesch, 2013; Sundell and Saylor, 2017). Mixture modeling is capable of more rigorously determining the magnitude of contribution of known and unknown sources to basin sediments (Sundell and Saylor, 2017).

In the context of this study, areas producing large amounts of MP pollution would be considered sediment sources. MPs are emitted from these sources, transported downstream, and ultimately deposited in sinks (Rangel-Buitrago et al., 2026). MPs have this source-to-sink behavior in common with other types of sediment, which are eroded within catchements and deposited downstream (Rohais et al., 2024). Just as for other sediments, rivers are the primary transporters of MPs to the ocean (Goßmann et al., 2025). However, different waterways may have different assemblages of MPs (Fiore et al., 2022). Because MPs can deposit anywhere that other sediments deposit, MP sinks abound and are varied across the landscape (Rangel-Buitrago et al., 2026).

Questions and Objectives

Because plastics are a new addition to sedimentary systems (e.g. Rangel-Buitrago et al., 2026), their characteristics and concentrations have been described in numerous recent studies. The current literature focuses mainly on describing the color, shape, chemical makeup, or quantity of MPs in an area (e.g., Fiore et al., 2022; Hasenmueller and Ritter, 2024; Goßmann et al., 2025). The existing research has not used detrital methods common in geologic studies to quantitatively define source-sink relationships in MPs composition and concentration in river systems. Microplastics are different from DZs in that they are not used to obtain ages to be used in provenance studies. However, the compositional data from MP samples may be able to be used in a similar way. Recent advancements in

imagery and multivariate analysis include the ability to determine MP chemical composition (Smolen et al., 2025). The data from these analyses can be made into probability distributions, which can be quantitatively compared like probability distributions of detrital zircon data. If detrital methods carry over into MPs research, then these source-sink relationships can be quantitatively and spatially defined.

In this study, we will apply these methods to characterize the residency distance of MP fingerprints, determine source-sink relationships, and establish baseline quantitative methods for MPs research. More specifically, our questions are (1) whether different tributaries to the Columbia River have different MP fingerprints (i.e., distinct suites of MP types), (2) to what degree the fingerprints of upstream/tributary sources affect downstream MP quantities and assemblages, and (3) whether MP assemblages record the land use of their source catchments. To address these questions, we will collect sand samples that likely contain MPs in areas of sediment deposition along the Columbia River and its major tributaries. We will apply methods rooted in MP chemistry to identify plastic types. We will then quantitatively compare our MPs samples using established sediment provenance methods used in geologic research to understand transport and distribution of MPs.

METHODS

Do different rivers exhibit distinct MP fingerprints?

In this study we test three hypotheses. The first is that different rivers exhibit distinct MP fingerprints. This is based on the premise that different watersheds are unique in the types of plastic that are found within them. These plastics will erode and then be found in waterways draining that catchment. To test the first hypothesis, we will employ a standard sample collection and separation strategy, follow procedures for processing samples with micro Fourier infrared (μFTIR) spectroscopy, analyze them with neural networks trained to identify and classify MPs, and use well-established sediment provenance methods of sample comparison and multidimensional scaling to visualize microplastics fingerprints across the CRW.

Pre-sampling Analysis

Before sampling, target areas of high MP pollution potential and MP contamination risk were determined for the CRW. All of the data collected in this part of the study was obtained from publicly-accessible databases. The data were then processed in Python. The level-6 hydrological sub-catchments for the CRW were obtained from the HydroBASINS database (Lehner and Grill, 2013). The pollution potential and risk contamination metrics were determined using population, land use, climate, and streamflow data. The population data were obtained from the US and Canadian censuses. These data were available by county, and so they were modified to apply to the sub-catchments. County populations for the US were retrieved via the US Census API, and county populations for British Columbia were obtained from a downloaded population shapefile (Statistics Canada, undated; U.S. Census Bureau, undated). The population data were then spatially overlaid for each sub-catchment so that each could have a population density attribute. A raster of land use data was downloaded from ESA WorldCover (European Space Agency, 2021). This data contained the categories of built-up (urban), cropland (agricultural), water, and the 5 categories that were simplified as wilderness in this study: forest, shrubland, grassland, bare/sparse, and snow/ice. The land use data were clipped to each sub-catchment, and land use percentage zonal statistics were calculated for each sub-catchment. WorldClim precipitation data were also available in a raster format (Fick and Hijmans, undated). Similarly to land use, these data were interpolated for the sub-catchments. Stream gages in the CRW were retrieved from the NWIS API (US Geological Survey, undated). The data from these gages were not used in any of the calculations, but rather as an optional dataset to include in future maps and calculations. The amount of streamflow generated in a sub-catchment was instead calculated as the product of precipitation, sub-catchment size, and runoff (a coefficient based on land use). This attribute was calculated for each sub-catchment.

For each sub-catchment within the CRW, potential pollution and risk of MP contamination were calculated. These metrics are based on the data available for the different areas of the watershed, and

there has been no ground-truthing process to assess their merit. The metrics do not represent the actual pollution or MP contamination hazard. However, they are useful in that they provide a starting place to help narrow down areas of sampling priority. The pollution potential metric is as follows:

$$P_{\varphi} = A_u \cdot \rho_p + A_a$$

where P_{φ} is pollution potential, A_u is urban area in km^2, ρ_p is the population density in people/km^2, and A_a is agricultural area in km^2.

The contamination risk metric is as follows:

$$R = (P_l + P_u) / Q_i \cdot C$$

where R is the risk of contamination, P_l is local pollution potential, P_u is upstream pollution potential, Q_i is normalized streamflow, and C is the runoff coefficient.

This preliminary analysis revealed that the catchments with the highest potential for MP pollution are those containing cities like Portland, Salem, Eugene, Vancouver, Yakima, Spokane, Coeur d'Alene, Boise, and Twin Falls (Figure 3). They are also the catchments containing large agricultural areas such as the Snake River Plain, the Treasure Valley, the Willamette Valley, and the Palouse. The catchments with the highest risk of MP contamination are those that are downstream of the highest potential polluters. These include areas such as the Tri-Cities on the eastern side of the Oregon-Washington border, Ontario, Oregon, which is just downstream of the Treasure Valley, and the part of the Cascades through which the Columbia River flows before it joins the Pacific Ocean. Though the ocean was not a part of this analysis, it undoubtedly receives a large quantity of MPs from the CRW.

Sampling Strategy

MP samples will be collected from streams across the CRW. The streams that are sampled, as well as the number of samples collected per stream, will be determined by the preliminary analysis. They may also be determined subjectively by factors such as magnitude of streamflow, proximity to Pocatello, and perceived agricultural, ecological, or economic importance. Though some areas of the CRW will be prioritized, time and resources may allow for a more complete characterization of the

entire watershed. Several samples will be taken along the main trunks of the Snake and Columbia Rivers, especially immediately after large urban and agricultural centers. Tributaries draining such urban and agricultural centers will also be prioritized or sampling. In sampling tributaries to the main rivers, three things must be characterized: the baseline MP fingerprint of the main river before the tributary, the MP fingerprint of the tributary itself, and the MP fingerprint of the main river after the MP load of the tributary has mixed with that of the main river. At least three sediment samples must be taken to characterize each one of these areas.

Sample collection

MPs will be sampled following standard sediment sampling methods outlined by Masura et al. (2015). Samples are taken from the downstream ends of point bars or mid-channel bars. Fine-grain, low-density sediments like MPs are primarily deposited in these areas due to hydraulic sorting (Rittenhouse, 1943). Sampling is done using plastic-free tools to avoid biasing MP concentrations. A metal shovel is used to collect from the top 1-2 cm of sediment. This will ensure that the sample represents recent MP deposition. Sediment samples will be stored in a 100% cotton bag to prevent contamination.

Sample separation

Before any separation is done, MPs must be thoroughly air-dried. Sediment will then be sieved at 1 mm. Of the sieved sediment, 150 g will be added to a 500 mL plastic centrifuge bottle with 300 mL of lithium metatungstate. Following methods from Grause et al. (2022), the bottle will be lightly agitated and spun in a centrifuge at 6000 RPM for 15 minutes. After this, the microplastics will be at the top of the solution and the denser sediment will be at the bottom. Using a custom trap-door-like device, the dense, non-plastic sediment will be discarded, and the remaining microplastics will be decanted into a vacuum funnel with an anodisc filter. At the end of filtration, the MPs will be resting atop the filter.

Compositional Analysis

Several steps will be taken in order to determine the chemical composition of each MP particle in a sample. First, the filter containing the MPs will be dyed with 4-dimethylamino-4’-nitrostilbene (DANS), which will allow for identification of plastic type based on fluorescence color. The filter will then be fluoresced and imaged under a micro Fourier transform infrared microscope. The resulting image will be processed by a convolutional neural network trained to detect MPs (Smolen et al., 2025). If an area on the image is flagged as containing MPs, it will be imaged at higher resolution in reflectance and transmittance modes. These images will be fed into a different neural network trained to de-noise the μFTIR spectra of MPs into attenuated total-reflectance (ATR) spectra. These spectra will then be processed by other neural networks, such as library matching (Cowger et al., 2021) and machine learning (Lin et al., 2022) methods, to ultimately determine polymer type for each MP particle (Smolen et al., 2025).

Sample Comparison

Sample comparison starts with making a histogram of the frequency of each plastic type in a sample. These compositional data will then be computed into distributions using a probability density function. We expect that these histograms and distributions will be unique for each sample and will represent the MP fingerprints of the area where the sample was collected (Figure 4). Sample distributions can be compared to each other using metrics like cross-correlation (Saylor et al., 2013), Likeness (Satkoski et al., 2013), Similarity (Bhattacharya, 1943), the Kolmogorov-Smirnov Test D statistic (Massey, 1951), the Kuiper Test V statistic (Kuiper, 1960), and others (Figure 5). A sample comparison matrix will be created by comparing several samples to one another. This matrix will undergo multidimensional scaling (MDS) ordination, which will transform and visualize the matrix in a plot that will show how dissimilar samples are by their distance in Cartesian space (Vermeesch, 2013) (Figure 6). MDS plots can play a useful role in interpretation. In this case, samples that plot closer together will have similar MP fingerprints.

Do downstream samples of MPs represent the admixture of upstream tributary sources?

The second hypothesis in this study is that downstream samples of microplastics represent the admixture of upstream tributary sources. This will be tested by running an inverse Monte-Carlo (top-down) model to characterize downstream samples in terms of percentage of microplastics contributed by upstream samples. The interpretive value of quantitative comparison described above can be taken further by using the MPs samples as parameters to top-down mixture models, which will quantify the upstream contribution to a downstream area (Sundell and Saylor, 2017). An inverse Monte-Carlo unmixing model takes one downstream sample distribution and several upstream sample (usually tributaries flowing in before the downstream sample) distributions as parameters. The model runs many iterations (usually 10,000) to quantify the percent contribution of MPs that each tributary sample contributes to the downstream sample. During each iteration, the algorithm compares the probability distribution of a randomly weighted aggregate of upstream/tributary samples to that of the downstream sample. The iterations that produce upstream/tributary sample aggregates most similar to the downstream sample are averaged to suggest the MP contribution percentage of each upstream area or tributary to the downstream area.

Although inherently arbitrary as all mixture modeling results are relative, if the model converges with an R^2 value greater than 0.7, then we consider that it has run successfully, and the downstream sample is faithfully derived from (and reflects the magnitude of) emitters in upstream tributaries. If it does not converge, then either the downstream samples do not represent the admixture of upstream sources, or one or many MP emitters are not represented in the upstream samples (Figure 7).

Do downstream MP assemblages record the land use of their source catchments?

Samples will be further compared through a bottom-up unmixing using non-negative matrix factorization (NMF), a collection of algorithms widely used in the unsupervised processing of large amounts of data, originally with applications in facial recognition (Lee and Seung, 1999) and more

recently in sediment provenance studies (Sharman and Johnstone, 2017; Saylor et al., 2019; Saylor and Sundell, 2021). Using only data from downstream samples, NMF can characterize likely upstream sources and their relative proportions by returning the weightings of probability distributions of those sources. These likely sources may be either geographic sources or land-use types, depending on how distinct the tributary fingerprints are, which can be evaluated by quantitatively comparing them with those of either empirical upstream sources or compound sources aggregated based on catchment land use. If the model returns the same number of likely sources as there are land uses drained by the tributaries, it would highlight the importance of land use in the fingerprint-forming mixture of microplastics downstream (Figure 8).

ANTICIPATED RESULTS

If different rivers exhibit distinct MP fingerprints, then samples from the same river will be more similar, and samples from different rivers will be more different from each other. On an MDS plot, the similar samples will cluster and the different ones will not (Figure 9). Samples may cluster based on upstream land use or geographic location. Streams may differ in MP composition because of differences in the ways that MPs enter the waterway. For example, a suburban area may contribute overwhelming amounts of polyester to the watershed because its inhabitants wash clothes made of polyester in water that becomes incorporated to the river. An agricultural area may contribute relatively high amounts of polyethylene, which is used in tarps, to the fingerprint. These different types of MPs in a sample will lead to different modes in the distribution. This will, in turn, influence the inter-sample comparison, and ultimately, the way that samples cluster on the MDS plot. If streams share the entry points for these types and quantities of MPs, they will cluster (Figure 9C). However, if MP fingerprints from different streams are not distinct, then MP samples will appear scattered (Figure 9D).

If downstream samples of microplastics represent the admixture of upstream tributary sources, then a downstream sample can be reliably reconstructed by a combination of upstream sources. If those samples are used as parameters in the top-down unmixing model, and if it converges on a reliable

reconstruction (e.g. $R^2 > 0.7$), then that downstream sample is indeed composed primarily of upstream sources (Figure 10). In smaller catchments where complete characterization is possible, this mixture model can provide meaningful results. However, in a larger watershed where not all tributaries can be realistically sampled, this model may suggest that the sampled tributaries have less of an effect on downstream areas than they really do. In smaller catchments, this model could perform well due to the ease of tributary characterization. However, in larger catchments where not all the tributaries can be characterized, the top-down mixture model may not converge with a reasonable R^2 value, even if the tributaries are contributing to the overall MP fingerprint.

Because different land uses may produce different MPs assemblages in a waterway, these land uses may be traceable upstream. When several downstream samples of an area are used as parameters in the bottom-up umixing model, the model will attempt to generate the likely upstream sources. If the model returns likely sources that reflect upstream land usages and/or geographical tributaries, then the downstream MP assemblages record the land use of their source catchments. On a small scale with a few distinct tributaries, the algorithm may return the geographical sources. On the other hand, on the large scale, it may return land usages of the catchment. If the model results do not match either catchment land use or upstream tributaries, then downstream MP assemblages do not reliably record land use, and/or there are other factors at play than just upstream land use (Figure 11).

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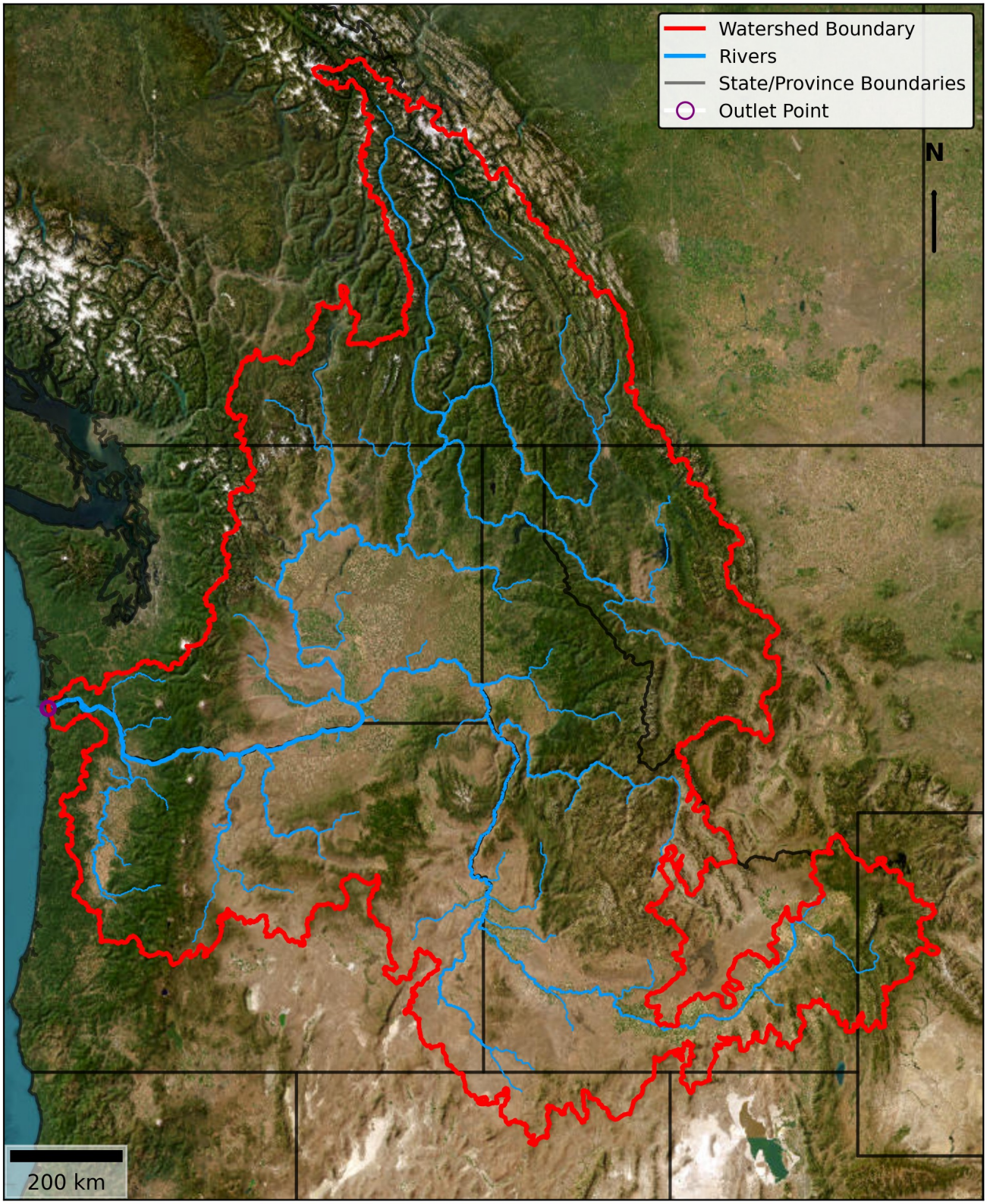
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386 **Figure 1**: Columbia River watershed with 5th to 9th order streams



387 **Figure 2a**: Land use across the Columbia River watershed. Urban land use is population-weighted so

388 that it can be seen on the map.

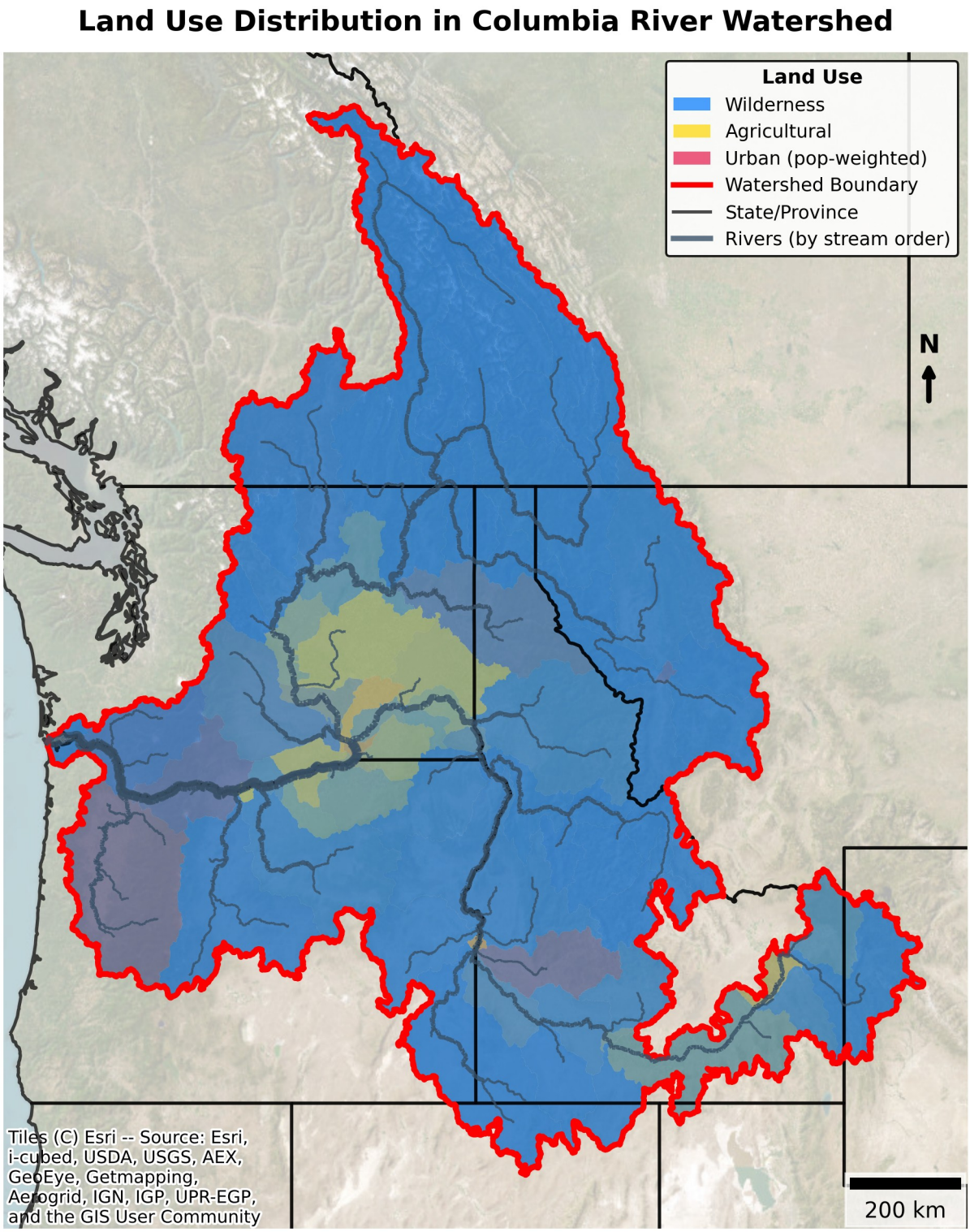
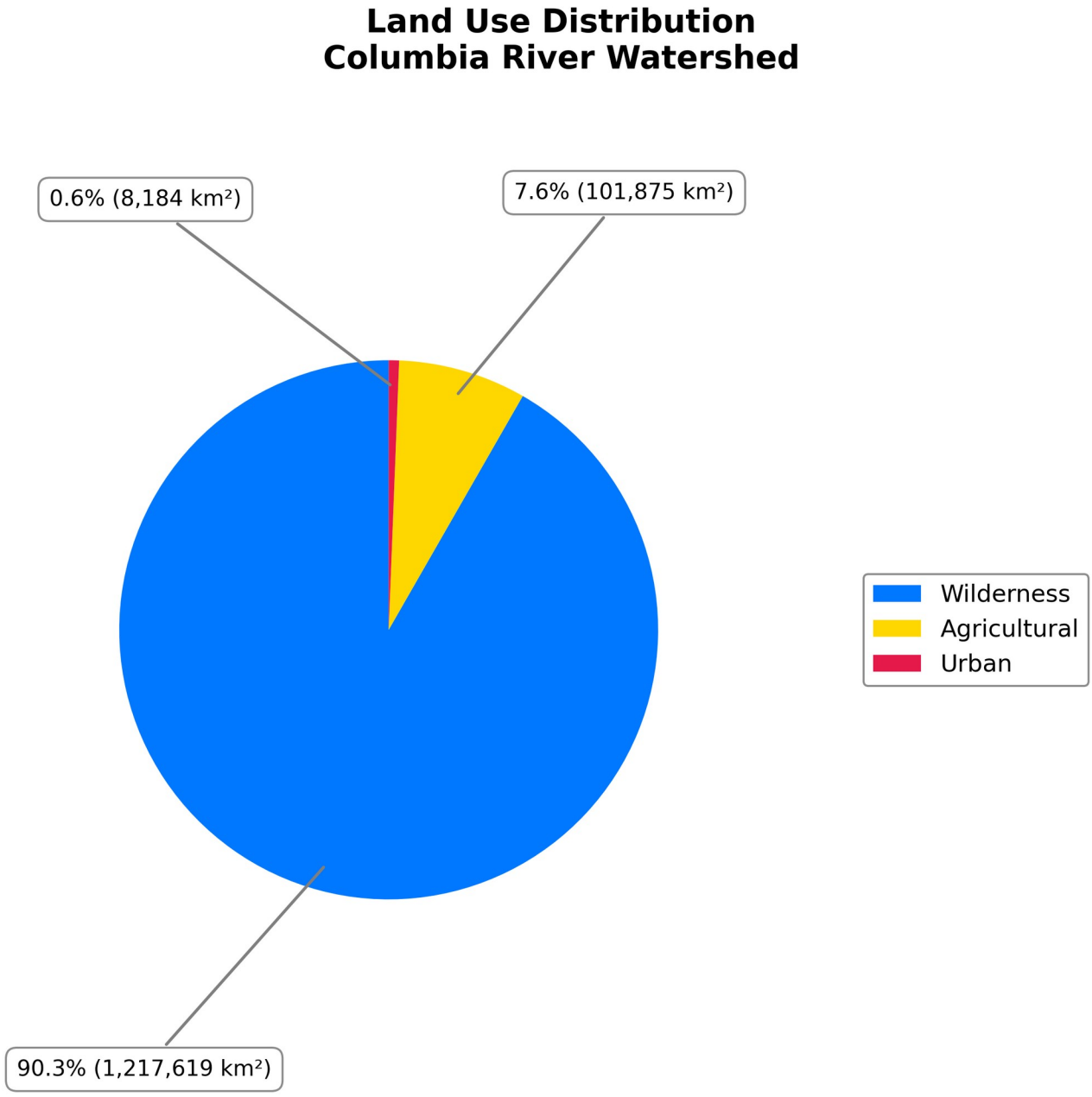
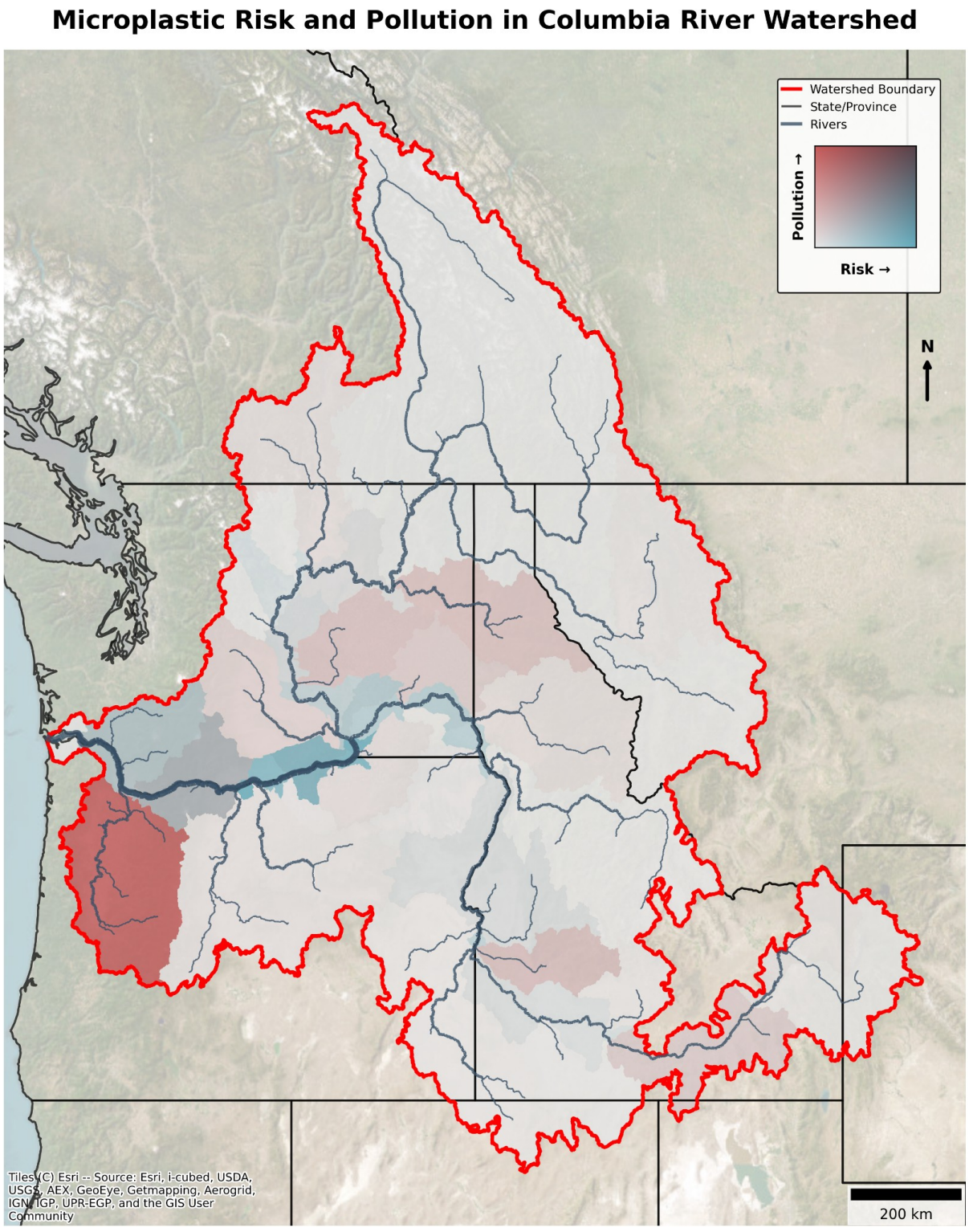


Figure 2b: Land use across the Columbia River watershed. The pie chart shows the actual proportions.



394 **Figure 3**: Results of preliminary MP pollution potential and MP contamination risk analyses.



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Figure 4: Vectorization workflow

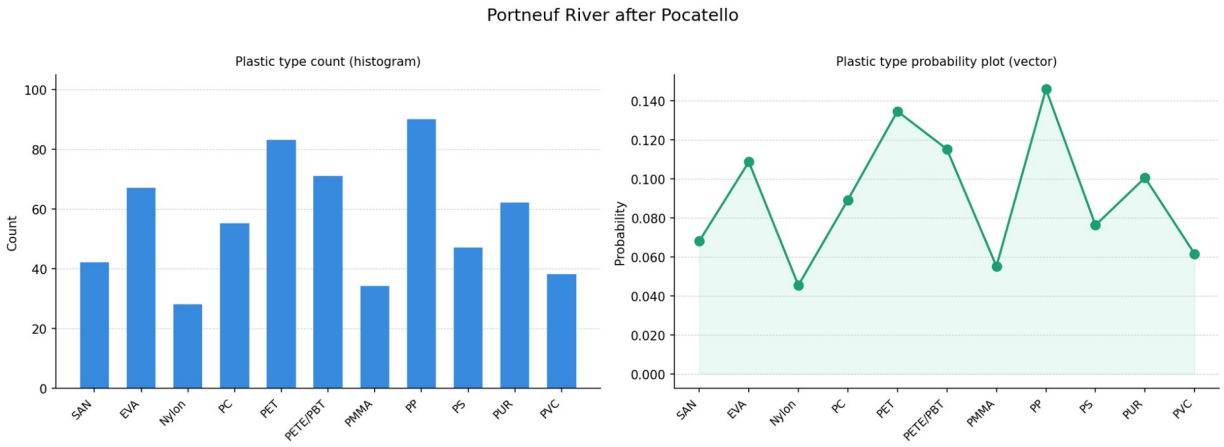
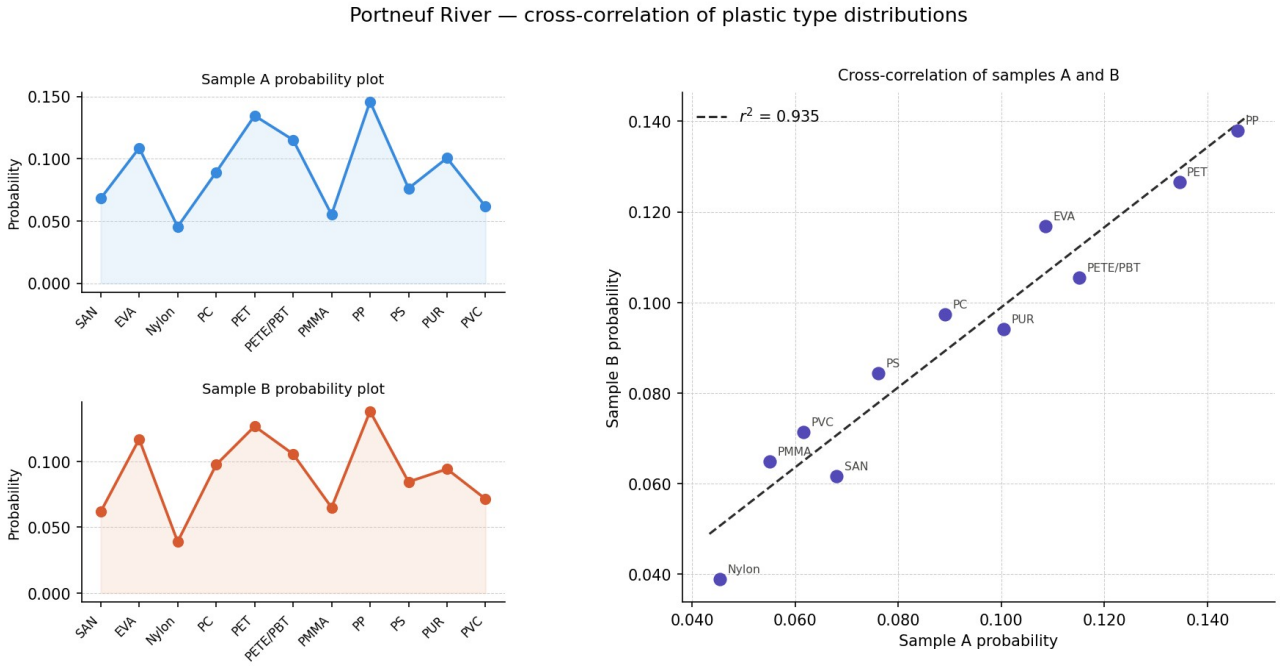
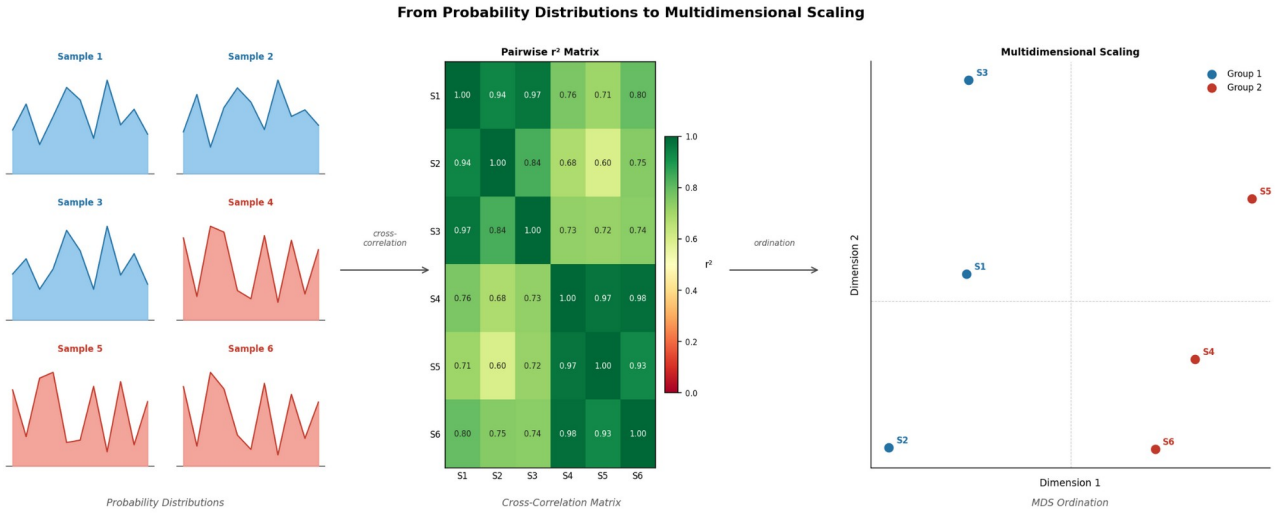


Figure 5: Quantitative comparison workflow



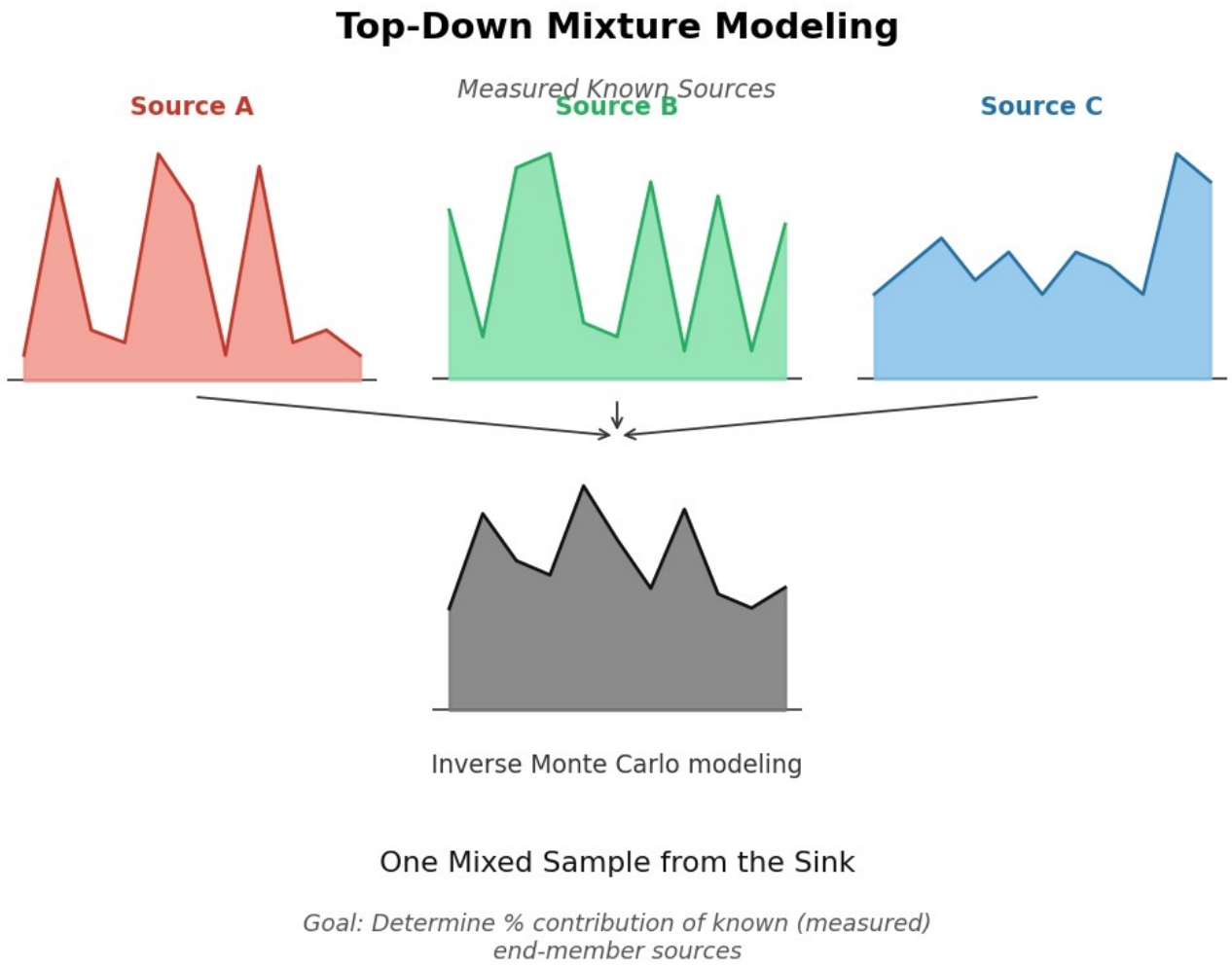
403 Figure 6: Multidimensional Scaling Ordination workflow



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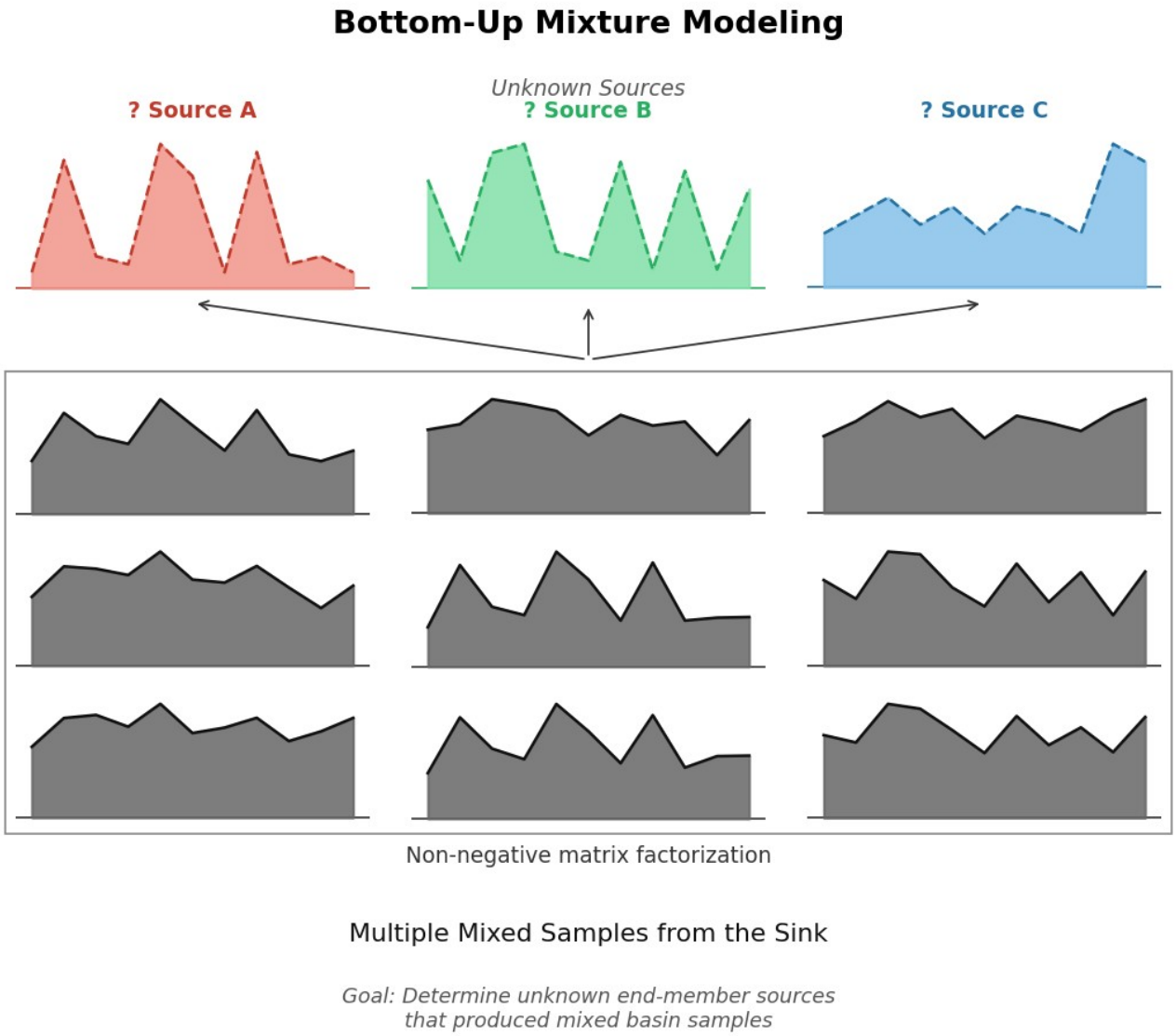
406 Figure 7: Top-down unmixing model workflow



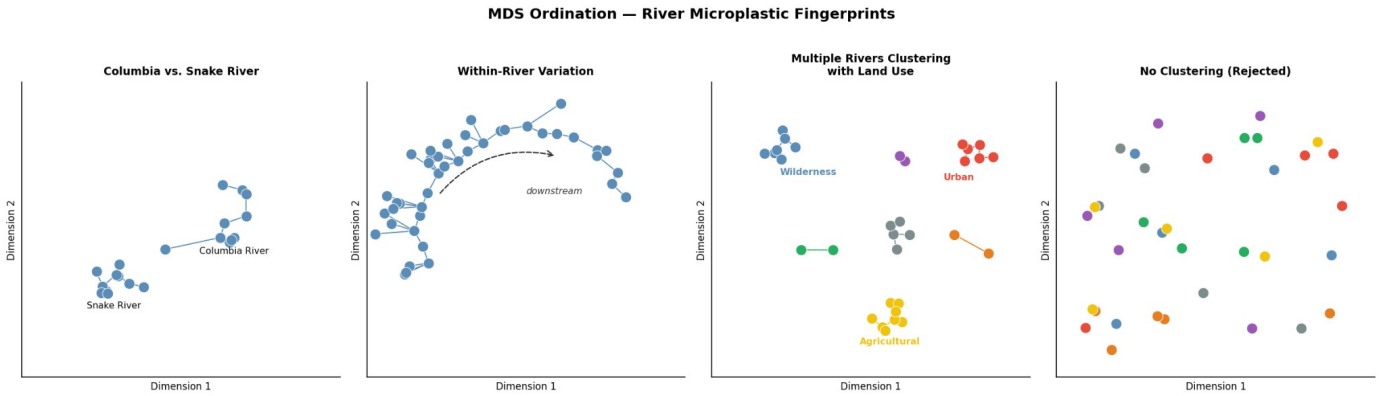
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Figure 8: Bottom-up mixing model workflow



411 **Figure 9**: Multidimensional scaling plots showing H_1 supported (left 3 plots) and refuted (right)



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Figure 10: Probability density plots showing H_2 supported (left) and refuted (right)

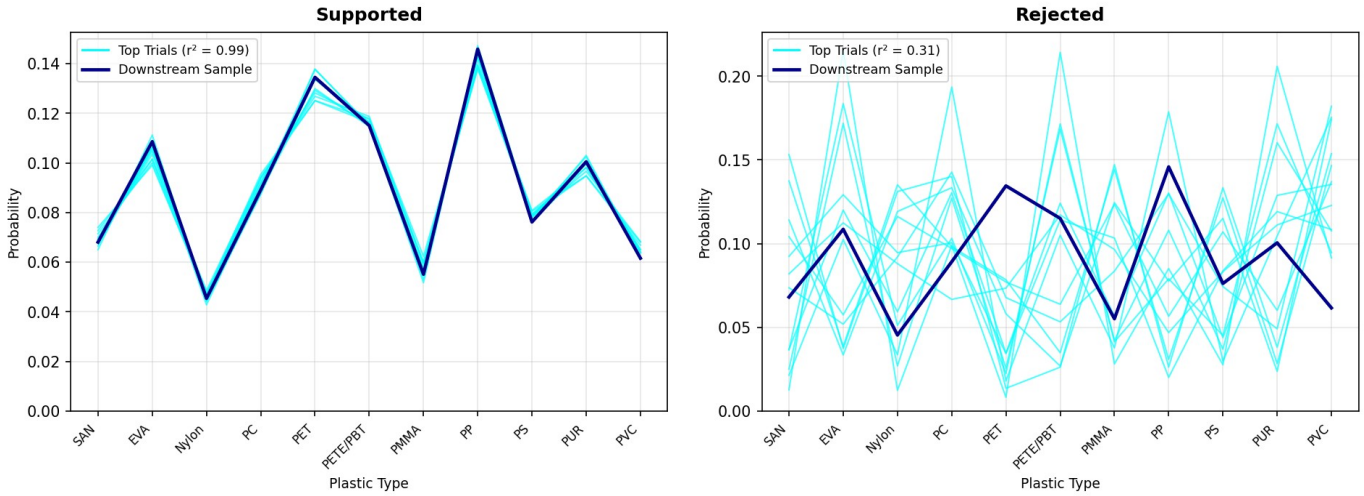


Figure 11: Probability density plots showing H_3 supported (left) and refuted (right)

